

Airborne Lead and Birth Outcomes: Evidence from Leaded Fuel in Auto Racing

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Abstract

The National Association for Stock Car Auto Racing (NASCAR) and Automobile Racing Club of America (ARCA) races continued to use leaded gasoline until 2007, despite the removal of leaded gasoline from on-road vehicles long ago. Exploiting this novel quasi experiment, I study how airborne lead exposure affected birth outcomes using administrative birth records linked to race events at Texas Motor Speedway (TMS) in Fort Worth, Texas. Mothers living within 25 km of TMS see a reduction in infant birthweight, shortened gestation length, and an increased probability of low birthweight before the ban of lead from NASCAR and ARCA races, while a post-ban placebo yields coefficients near zero. To translate race-day exposures into blood lead burdens, I developed a pregnancy-specific physiologically based pharmacokinetic (PBPK) model that simulates maternal and fetal daily blood lead trajectories. I find that a one-standard-deviation increase in second-trimester PBPK blood lead reduces birthweight by approximately 63 grams. A potential mechanism suggests that lead increases gestational hypertension and shortens gestation, and these effects disappear after the transition to unleaded fuel. These findings suggest that even a relatively low level of lead can have a meaningful effect on maternal and infant health.

JEL classifications:

Keywords: Lead exposure, birthweight, prenatal health, NASCAR, leaded gasoline, PBPK model

1 Introduction

Lead (Pb) is one of the most well-established environmental neurotoxins and has been associated with numerous adverse outcomes, including cognitive impairment, increased criminal behavior, and higher infant mortality. It is found in many places such as small piston engine aircraft, older housing paint, and drinking water. Growing evidence on the harmful effects of lead prompted many countries to adopt policies aimed at reducing exposure beginning in the 1970s. One of the most consequential interventions was the phaseout of leaded gasoline for passenger vehicles, which produced substantial public health gains. However, the National Association for Stock Car Auto Racing (NASCAR) and the Automobile Racing Club of America (ARCA), two of the largest racing organizations in the US, continued to use leaded racing fuel until 2007 before switching to unleaded gasoline. These localized sources of airborne lead continue to generate meaningful health costs, and the burden falls on vulnerable populations, including pregnant women and developing fetuses.

Exploiting this quasi-experimental setting where NASCAR and ARCA generated concentrated bursts of airborne lead emissions surrounding the racetracks before the switch in 2007, I estimate the effect of prenatal lead exposure on birth outcomes among pregnant mothers living near Texas Motor Speedway (TMS), which is by far the most active NASCAR and ARCA venue in Texas during the sample period (2003-2010). I identify the causal effect of prenatal lead exposure entirely from *within-TMS variation* as cross-track design poses unreliable estimates. The within-TMS design exploits four sources of variation in exposure among mothers living near the same racetrack. First, mothers living closer to TMS are likely to experience higher lead exposure as lead concentrations decline with distance from the track. Second, wind direction on race days is plausibly exogenous to maternal characteristics and/or sorting, and hence, mothers who reside downwind of the lead plume should expect to see a higher birthweight reduction than mothers who are live outside the lead plume. Third, the gestational timing of each race depends on the mother's conception date. Mothers who live the same distance to TMS but conceive just a few weeks may experience different patterns of race exposure across trimesters. Finally, emissions vary across races with caution flags, accidents, and race length which generates variation in the amount of lead released that is plausibly unrelated to maternal health.

By focusing on mothers living near TMS, the within-track design also avoids the concerns associated with cross-track comparisons in which mothers living near active racetracks may differ systematically from those living near inactive racetracks. Because all mothers in the analysis sample live within 25 kilometers of TMS, identification comes from the four sources of variation described above. Meyer Speedway in Houston and Texas

World Speedway in College Station are two legacy racetracks that had no NASCAR and ARCA races during the sample period. Meyer Speedway showed a significant decline in birthweight of approximately 25 grams post 2007 which serves as an unreliable venue for cross-track comparison design. Texas World Speedway, however, exhibits no analogous decline which could imply that there is an artifact in the Houston metropolitan area that play a role. The last of the four racetrack venues in Texas, Thunder Hill Raceway in Kyle, hosted five K&N Pro Series West (KNPSW) regional races between 2005 and 2009 that generated a lower-intensity emission, and the sample is too small and noisy to serve as an independent replication.

To study the effect of prenatal lead exposure on birth outcomes, I first determined how much lead was emitted in each race and then summed over all races that occurred in each trimester to construct a trimester-level exposure measure. Then, I ran a split-sample regression comparing births before and after the 2007 ban on leaded gasoline in race cars. This approach has a flavor of a difference-in-differences design since identification comes from comparing the pre- and post-ban slopes. The identifying assumption is the same as in any DiD: factors correlated with race-day intensity (i.e., traffic, noise, pollution) are present in both periods. Under this assumption, such confounds should bias the pre- and post-ban coefficients in the same way. If, however, the pre-ban coefficients are negative and significant while the post-ban coefficients are zero, then this is evidence of lead exposure, as lead is the only thing that is present pre-ban and absent post-ban. While it is straightforward to model the effect through a simple reduced-form equation, it is, however, more complex and often not accounted for in many epidemiological studies how lead deposits and accumulates within the body, especially in pregnant women. Lead could be deposited, accumulated, and remobilized in her body across trimesters. To understand how lead behaves during pregnancy, I developed a pregnancy-specific Physiologically-Based Pharmacokinetic (PBPK) model that translates each mother's discrete race-day exposure pulses into daily blood lead trajectories over a 280-day pregnancy. The model compartmentalizes lead in the maternal blood, tissues, and bones and return trimester specific blood lead means, area under the curve (AUC) and peak burdens while capturing the re-mobilization of lead stored in maternal bones during her later trimesters. I find that the second and third trimesters see a reduction in birthweight, a higher probability of low birthweight, and a decrease in gestational age. Births conceived after the ban see no effect. This suggests that the pre-ban effect is driven by lead rather than an artifact of race-day activity (e.g., traffic). A possible mechanism suggests that lead exposure raises the probability of gestational hypertension, shortens gestation, and reduces birthweight, while the effect is null post-ban. The effects are strongest for mothers residing closer to the racetrack and downwind on race days.

A large body of work documents that exposure to ambient pollution reduces birth weight, increases the risk of preterm birth, and worsens a wide range of other infant health outcomes (Pedersen et al., 2013; Stieb et al., 2012; Currie and Neidell, 2005; Currie, 2013). Lead exposure, in particular, is found to harm cognitive development, reduce educational attainment, increase criminal behavior, lower adult earnings, and generate persistent socioeconomic disadvantages over the life cycle (Aizer et al., 2018; Jenkins et al., 2022; Liu et al., 2014; Talayero et al., 2023; Boyle et al., 2021). However, most of this literature faces challenges in isolating the specific source of pollution exposure and measuring the timing of exposure during pregnancy. Less is known about prenatal exposure to airborne lead, despite the fact that lead readily crosses the placenta and can affect fetal development in utero (Zhu et al., 2010; Yüksel et al., 2022; Santana et al., 2023). This chapter makes four contributions. First, it provides causal evidence from a within-track design that localized sources of airborne lead continue to generate measurable maternal and fetal health costs even in a low-background pollution environment. Second, it is the first paper, to the best of my knowledge, that introduces a biologically grounded PBPK framework into the empirical environmental-health literature on lead exposure and birth outcomes and shows which gestational windows are most harmed by lead. Third, the unit of analysis is conducted at a more precise, neighborhood-level scale than in previous studies, which have a coarser definition of exposure at the county level. Finally, it accounts for wind direction, allowing me to more precisely identify mothers who were exposed downwind on race days rather than defining exposure based solely on proximity to the racetrack.

The remainder of the paper proceeds as follows. Section 2 provides background on leaded gasoline and NASCAR's and ARCA's switch to unleaded gasoline in 2007. It also introduces the four Texas racetracks used in the analysis, with Texas Motor Speedway (TMS) as the primary track. Section 3 describes the birth records, race data, and weather data. Section 4 describes how I construct race-level lead exposure measures, along with the PBPK model used to trace lead exposure over the course of pregnancy. Section 5 describes the empirical strategy, which uses variation in the timing of exposure, distance from TMS, and wind direction. Section 6 presents the main results and examines heterogeneity by distance and wind direction, the PBPK estimates, and maternal hypertension as a potential mechanism. Section 7 discusses the magnitude of the estimates and concludes. Finally, the appendix contains the robustness and falsification tests and monitor-based exposure measures.

Texas NASCAR racetracks and maternal residential locations
Live births within 25 km of each track (2003--2010); dashed ring = 25 km

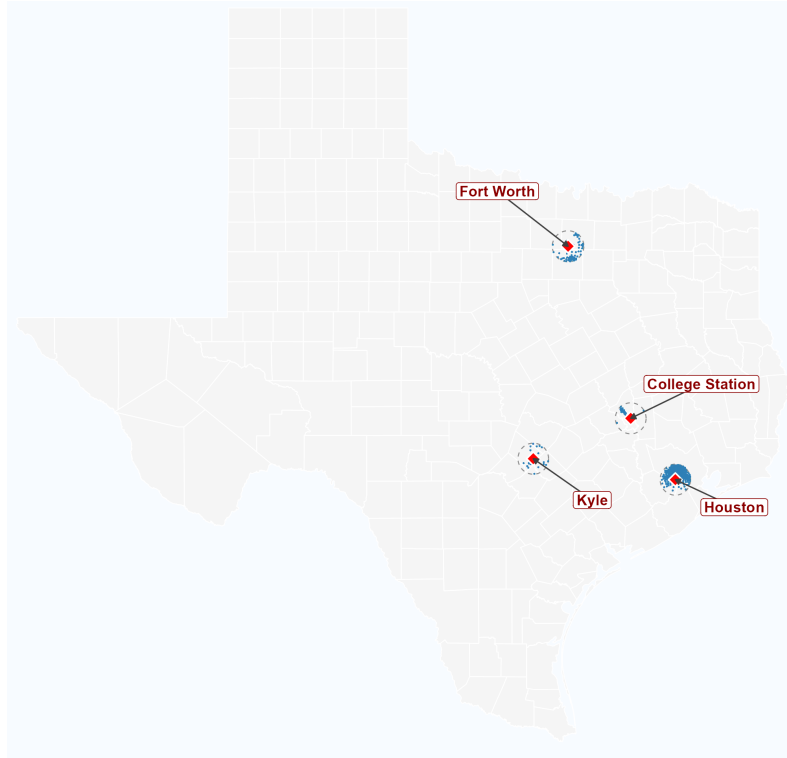


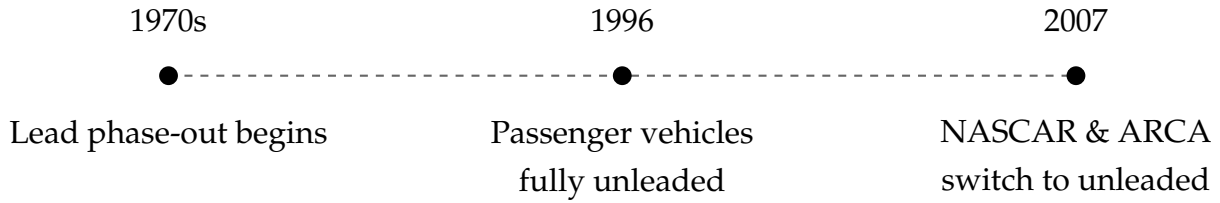
Figure 1. Red diamonds indicate the four racetracks in Texas located in Fort Worth, College Station, Kyle, and Houston. Blue circles indicate maternal residential locations (census tract centroid) in the birth sample.

2 Background and Institutional Setting

2.1 Leaded Fuel in NASCAR & ARCA Races

Leaded gasoline remained common in motorsports long after it disappeared from commercial passenger vehicles. Lead served as an octane booster which allows engines to operate at very high compression without engine knockoff. Prior to 2007, NASCAR and ARCA races used fuel containing approximately 4 to 5.2 grams of tetraethyllead per gallon ([Hollingsworth et al., 2025](#)). The organizations operate hundreds of races each year through both national and regional racing series. NASCAR's three national series are the Cup Series, Xfinity Series (currently called O'Reilly Auto Parts Series), and Camping World Truck Series (currently called CRAFTSMAN Truck Series) while ARCA's main national series is the Menards Series. NASCAR also hosted some regional series, namely, K&N Pro Series (East and West), which operated in the Northeastern and Mid-Atlantic regions and West Coast and throughout the Rocky Mountain region in the United States. These races follow a strict guideline on the type of vehicles and fuel used in each race. NASCAR and ARCA required all cars in a given series to use the same standardized fuel

supplied by its fuel sponsor, Sunoco ([Hollingsworth and Rudik, 2021](#)). This standardization implies that variation in lead exposure arises from the timing and location of races rather than from differences in fuel composition across drivers or teams.



The Clean Air Act of 1970 established the federal framework for regulating air pollution in the United States. Beginning in the 1970s, the EPA introduced regulations to gradually reduce lead in gasoline, in part because of growing concerns about the health effects of lead exposure. By 1996, leaded gasoline was completely phased out for on-road vehicles in the United States. Racing vehicles, however, were exempt from these requirements and continued to use leaded gasoline in racing cars. NASCAR and ARCA began switching to unleaded gasoline in 2006 and permanently switched their national racing series in 2007. NASCAR’s Xfinity and Camping World Truck Series began deleading on September 23, 2006 while the Cup Series continued using leaded gasoline through the end of the 2006 season and made its permanent switch on February 25, 2007. ARCA also made the switch starting in 2007¹. Given this staggered timeframe, I categorized the sample into three windows: pre-ban, transitional, and post-ban. The pre-ban window includes births that were conceived before December 17, 2005, so that their third trimester ended before September 23, 2006. The post-ban window contains births that were conceived on or after March 1, 2007, so that the entire pregnancy fell after the Cup Series switch on February 25, 2007². The transitional window includes births that were conceived between December 17, 2005, and February 28, 2007, and is excluded from the cross-sectional sample to prevent gestational overlap between pre and post ban windows.

2.2 Study Area

Texas Motor Speedway (TMS), located in Fort Worth, Texas, is the primary venue in this study. TMS is a 1.5-mile (2.41-km) quad-oval racetrack situated on roughly 1,500 acres (about 2.3 square miles). The speedway opened in 1996, hosted its first major race in 1997, and has since become one of NASCAR’s most active national venues. During the

¹Throughout the sample, ARCA had only one race at TMS in 2010, and hence, the entire analysis rests on NASCAR races which I will mainly refer to throughout this paper.

²I set the post-ban cutoff at March 1, 2007 rather than February 26, 2007 as this provides additional buffer to guard against measurement error in conception date which is calculated from the imputed date of birth for the 2008-2010 births.

sample period (2003-2010), TMS typically hosted six NASCAR race weekends per year from the Cup Series, Xfinity Series, and Truck Series in April, June, and November.

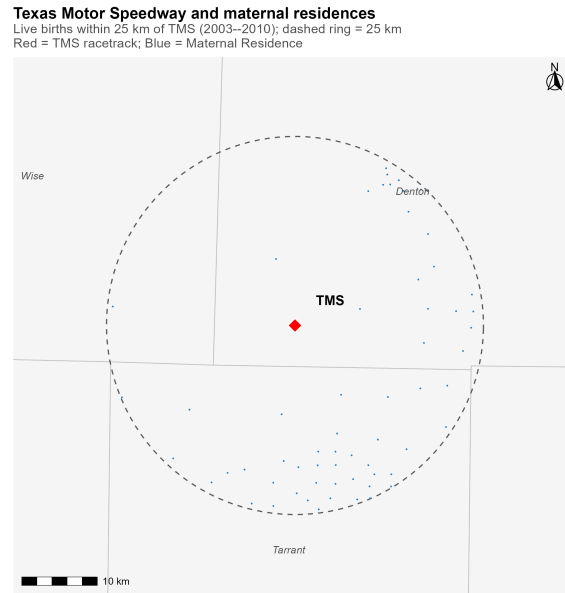


Figure 2. Figure 2 illustrates births within 25 km of TMS. The red diamond indicates the TMS racetrack located in Fort Worth. Blue circles indicate maternal residential locations (census tract centroid) in the birth sample. The sample contains approximately 320,000 births.

The second NASCAR's venue, Thunder Hill Raceway, is located in Kyle, Texas, and it hosted five races in the K&N Pro Series West (KNPSW) between 2005 and 2009. KNPSW switched to unleaded gasoline in 2007 but emits only a small fraction of the lead per race than the national series do. Texas World Speedway, located in College Station, and Meyer Speedway, located in Houston, are former NASCAR tracks whose last races occurred decades before the sample period (1993 and 1971, respectively).

Table 1. All Racetracks in Texas

Stadium ID	Track Name	City	Lap Length	Period Active	Role
34	Texas Motor Speedway	Fort Worth	1.5 mi	1996–present	Primary
14	Thunder Hill Raceway ^a	Kyle	0.375 mi	1998–2009	Secondary leaded
18	Texas World Speedway	College Station	2.0 mi	1969–1993	Legacy track
13	Meyer Speedway	Houston	0.5 mi	1969–1971	Legacy track

^a Also known as Central Texas Speedway.

TMS hosted Cup Series, Xfinity Series, and Camping World Truck Series races during all years of the sample. Thunder Hill hosted only KNPSW (regional) races. TMS also hosted one ARCA's race in 2010. Texas World and Meyer hosted no NASCAR's events during the sample period.

3 Data

The analysis combines three primary data sources, which are linked at the individual birth level.

Birth records. I use confidential Texas birth records from 2003 to 2010, obtained from the Texas Department of State Health Services (DSHS) Vital Statistics. These records contain detailed information on exact birth dates and time, location, birth outcomes, maternal characteristics, pregnancy complications, and the census tract of maternal residence. The primary outcome is birthweight in grams. Secondary outcomes include gestational age in weeks, low birthweight status ($< 2,500$ grams), and birthweight conditional on gestational age. Because the confidential data identify maternal residence only at the census tract level, I assign exposure using the centroid of the maternal tract rather than the mother's exact address. In addition, the 2008–2010 birth records do not report the exact date of birth. For these years, I impute the delivery date using the mother's reported last menstrual period and gestational length. Because the data are provided for different years, I harmonized the variables across years. I also restricted gestation length to a plausible range of 21 to 44 weeks, restricted birthweight to between 500 and 6,000 grams, and limited the sample to singleton births.

Race data. Race data come from Racing Reference that contains every NASCAR's and ARCA's event held at the four Texas racetracks during the sample period: Texas Motor Speedway in Fort Worth, Meyer Speedway in Houston, Texas World Speedway in College Station, and Thunder Hill Raceway in Kyle. These data include the exact race date, track location, racing series, total laps, lap length, miles traveled, number of drivers, total race duration in seconds, and other race characteristics. The data distinguish across NASCAR's major racing series, including the Cup Series, Xfinity Series, and Truck Series, which is important because different series transitioned away from leaded fuel at different times. For example, NASCAR Xfinity and NASCAR Craftsman Truck Series (CWTS) switched permanently to unleaded fuel in September 23, 2006 while NASCAR Cup Series switched to unleaded fuel in February 25, 2007. The sample also includes one ARCA race at TMS in 2010, which used unleaded fuel.

Weather data. Weather data comes from the National Oceanic and Atmospheric Administration's (NOAA) NCEP-NCAR Reanalysis 1 model. Wind direction, wind speed, precipitation, air temperature, and relative humidity on each race day come from weather stations within 275 km of each track. Each station contains an exact latitude and longitude that could be mapped to the birth and race data. For each birth $\times$ race day, I match to the nearest weather station and compute the bearing from the racetrack to the maternal

census tract centroid. A mother is classified as “downwind” on race day j if the angle between the track-to-residence bearing and the observed wind direction is within 45° .

For each birth, I compute the conception date as the date of birth minus the length of pregnancy in weeks (multiplied by seven). Then, I assign every race in the sample period to a trimester relative to that pregnancy such that Trimester 1 (T1) covers conception through gestational day 97; T2 covers days 98-195; and T3 covers days 196-280, following ACOG conventions (ACOG, 2024). Each mother is matched to the nearest of the four Texas tracks; the analysis sample then restricts to mothers whose nearest track is TMS and who reside within 25 km of TMS. Hence, the unit of analysis is birth by race days.

Table 2. Summary statistics: TMS analysis sample (within 25 km, live births)

	Pre-ban		Post-ban		Full sample	
	mean	sd	mean	sd	mean	sd
Birthweight (g)	3358.6125	(518.4447)	3342.4662	(524.6415)	3349.8444	(522.9881)
Low birthweight (<2500g)	0.0462	(0.2099)	0.0501	(0.2181)	0.0486	(0.2150)
Gestational age (weeks)	38.6518	(2.0521)	38.5895	(2.0516)	38.6065	(2.0422)
Apgar score (5 min)	8.9118	(0.5212)	8.8869	(0.6087)	8.8995	(0.5718)
Gestational hypertension	0.0427	(0.2021)	0.0431	(0.2030)	0.0435	(0.2039)
Eclampsia	0.0013	(0.0363)	0.0006	(0.0237)	0.0009	(0.0293)
Gestational diabetes	0.0220	(0.1466)	0.0358	(0.1858)	0.0299	(0.1703)
Maternal age (years)	28.7391	(5.8357)	28.5403	(5.8829)	28.6166	(5.8570)
Married	0.8066	(0.3950)	0.7529	(0.4313)	0.7792	(0.4148)
Education	0.4085	(0.4916)	0.3861	(0.4869)	0.3949	(0.4888)
WIC enrollment	0.2448	(0.4300)	0.2673	(0.4425)	0.2579	(0.4375)
Black	0.0457	(0.2088)	0.0534	(0.2248)	0.0498	(0.2175)
White	0.8751	(0.3306)	0.8539	(0.3532)	0.8606	(0.3464)
First birth	0.4043	(0.4908)	0.4094	(0.4917)	0.4079	(0.4914)
Insured	0.8878	(0.3156)	0.8901	(0.3128)	0.8863	(0.3174)
Distance to TMS (km)	19.1610	(4.5959)	18.7288	(4.9156)	18.9261	(4.7683)
Races during T1	1.1531	(1.0329)	1.6229	(1.2655)	1.4079	(1.1792)
Races during T2	1.2221	(1.0657)	1.5861	(1.2690)	1.4260	(1.1864)
Races during T3	1.0715	(1.0288)	1.4140	(1.2669)	1.2564	(1.1873)
Mean fetal blood lead, T1	0.1820	(0.2212)	0.0000	(0.0000)	0.1094	(0.1912)
Mean fetal blood lead, T2	0.3174	(0.2139)	0.0000	(0.0000)	0.1759	(0.2190)
Mean fetal blood lead, T3	0.3968	(0.2425)	0.0000	(0.0000)	0.2027	(0.2523)
AUC, full pregnancy	82.6630	(42.3315)	0.0000	(0.0000)	45.1854	(50.4848)
Peak blood lead	2.4123	(0.9255)	0.0000	(0.0000)	1.3083	(1.3339)
Observations	31141		28399		70847	

Notes: Mean with standard deviation in parentheses. Pre-ban is defined as conception before December 17, 2005; post-ban as conception on or after March 1, 2007. Full sample includes all births in the study area (including the transition window excluded from the pre/post columns). Sample consists of all Texas births from 2003–2010 whose nearest racetrack is TMS and who reside within 25 km. Education is defined as mothers with at least a college degree or higher.

4 Exposure Construction

For each birth-race pair, I computed total leaded emissions for each race and then summed the lead released during each trimester and over the full pregnancy to measure exposure. Because the amount of lead emitted depends on the lead content of the fuel, lead emissions after the 2007 ban are zero by construction. This provides a clean measure of exposure.

4.1 Emissions-Based Exposure

Total lead emitted at race j is calculated as:

$$\text{LeadEmitted}_j = \frac{\text{MilesTraveled}_j}{\text{MPG}} \times \text{LeadPerGallon} \quad (1)$$

where LeadEmitted_j is the total lead mass in grams released during race j ; MilesTraveled_j is the total miles driven by all cars during race j ; MPG is miles per gallon and is assumed to equal five, which is typical fuel economy for NASCAR; and LeadPerGallon is set to 5.2 grams, the average pre-ban tetraethyllead concentration per gallon ([Hollingsworth et al., 2025](#)). The formula is only meaningful for pre-ban races while post-ban calculation will serve as a counterfactual for what emissions would have been as if lead had still been used.

For each mother i and trimester $k \in \{1, 2, 3\}$, I sum the total amount of lead emitted from Equation 1 across all races at TMS whose date falls within trimester k of her pregnancy:

$$E_{ik} = \sum_{j \in \text{trimester } k(i)} \text{LeadEmitted}_j. \quad (2)$$

Because the distribution is right-skewed, I take the log plus one transformation

$$\tilde{E}_{ik} = \log(1 + E_{ik}). \quad (3)$$

where $\tilde{E}_{ik}$ is the log emission for mother i in trimester k used throughout the reduced-form analysis. Under the hypothesis that the pre-ban effect operates through lead, the post-ban coefficient on $\tilde{E}_{ik}$ should be near zero. However, if a post-ban coefficient is significant, this would indicate that non-lead factors, such as noise, traffic, or residential sorting near the track, may be driving the result. If so, attributing the pre-ban effect specifically to lead would be unwarranted.

4.2 Distance and Wind Direction

Distance and wind direction are two identifying sources of variation in this study. First, a mother who lives closer to the racetrack should experience higher exposure of lead than a mother who lives farther away. Therefore, birthweight should attenuate with distance from the racetrack. For each mother i , I compute distance, d_i , in kilometers from her census tract centroid to TMS and restrict the sample to mothers living within 25 km of TMS. Second, I classify mother i as living “downwind” of TMS on each race day if the angle between the bearing from TMS to her census tract centroid and the observed wind direction at the nearest weather station is within $\pm 45^\circ$.

For each trimester k of mother i ’s pregnancy, I construct two downwind measures. The first measure is the fraction of trimester k race days on which she was downwind of TMS,

$$\text{ShareDownwind}_{ik} = \frac{\#\{\text{trimester-}k \text{ race days with } i \text{ downwind}\}}{\#\{\text{trimester-}k \text{ race days}\}}, \quad (4)$$

defined as zero for mothers with no race days in trimester k and one for mothers who reside downwind on every trimester k race day. If mother i was downwind of the racetrack on two of three race days during trimester k , for instance, then her downwind exposure share would be 66 percent. The second is a binary measure,

$$\text{AnyDownwind}_{ik} = 1 [i \text{ is downwind on at least one trimester } k \text{ race day}], \quad (5)$$

captures whether a mother experienced any plume exposure during the trimester at all. If she resides downwind for at least one race day in trimester k , then she is considered exposed to the plume in that trimester. It is possible for mother i to reside downwind on all three of her trimesters using AnyDownwind_{ik} indicator.

Downwind classification on a race day

Mother A's bearing from TMS falls within the $\pm 45^\circ$ plume cone (downwind);
Mother B's bearing falls outside (upwind).

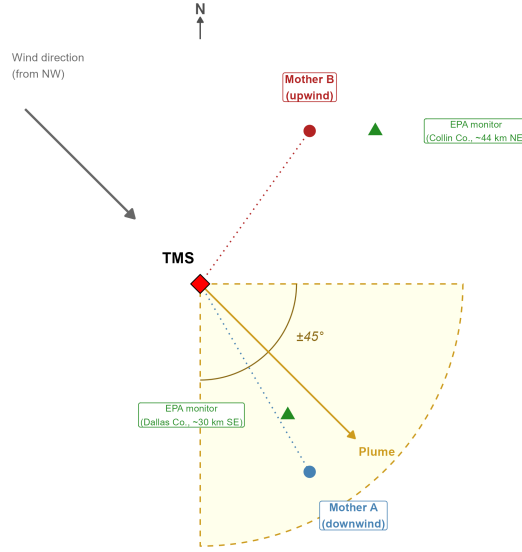


Figure 3. On each race day, a mother i is classified as residing “downwind” of TMS if the angle between (a) the bearing from TMS to her census-tract centroid and (b) the observed wind direction at the nearest weather station that day falls within $\pm 45^\circ$.

4.3 Pregnancy-Specific PBPK Model

A central challenge in this setting is that race-day lead exposure is intermittent. Because the half-life of lead in blood is approximately 30 to 40 days and weeks or decades for tissues and bone, exposure occurring late in one trimester can elevate maternal and fetal blood lead well into the next. Moreover, lead that is stored in maternal bone may be slowly re-mobilized back into the blood throughout the entire pregnancy (Télliez-Rojo et al., 2004). Hence, the standard reduced form in Equation 3 cannot capture these intra-pregnancy dynamics. It identifies whether a mother was exposed and the trimester in which exposure occurred, but it does not account for the accumulation and redistribution of lead across gestation.

To account for this, I develop a pregnancy-specific physiologically based pharmacokinetic (PBPK) model. The model follows the framework in O’Flaherty (1991); Oflaherty (1993, 1995) and represents the maternal body using three lead compartments: blood, soft tissue, and bone. When mother i is exposed to lead released during a race, the dose enters her blood as a discrete pulse on the gestational day on which the race occurs. I normalize the dose for each race, q_{ij} , using the race’s lead emissions from Equation 1. Because the dose is based on race-level emissions, all mothers exposed to the same race receive the same pulse in the PBPK model. Differences in exposure by distance from the track are

considered separately in the empirical analysis.

I compute the average lead emissions across all leaded race-days occurring during pregnancy in the full sample, and divide each race's emissions by this sample average. The resulting normalized dose measures the size of each race relative to an average race. For example, a value of 2 means that the race emitted twice as much lead as the average race. Race-day exposure is then represented as

$$E(t) = \sum_{j=1}^J q_{ij} \delta(t - t_j) \quad (6)$$

where $E(t)$ is the total lead input from race-day exposure on gestational day t ; J is the number of races during mother i 's pregnancy; q_{ij} is the normalized lead dose for race j ; t_j is the gestational day on which race j occurs; and $\delta(\cdot)$ is the Dirac delta function. Thus, each race contributes a single exposure pulse on the day it occurs.

Suppose $B(t)$, $S(t)$, and $K(t)$ denote lead in blood, soft tissue, and bone, respectively. The governing equations are

$$\dot{B}(t) = E(t) + k_{KB}(t)K(t) + k_{SB}S(t) - (k_{BS} + k_{BK} + k_{B0})B(t) \quad (7)$$

$$\dot{S}(t) = k_{BS}B(t) - k_{SB}S(t) \quad (8)$$

$$\dot{K}(t) = k_{BK}B(t) - k_{KB}(t)K(t) \quad (9)$$

Here, k_{B0} is the rate at which blood lead is eliminated through excretion; k_{BS} and k_{SB} are the rates of transfer between blood and soft tissue; and k_{BK} and $k_{KB}(t)$ are the rates of transfer between blood and bone. Take $\dot{B}(t)$ for example, the change in blood lead at a given time equals to lead entering the blood from exposure plus lead returning to the blood from bone and soft tissue minus lead leaving the blood for soft tissue, bone, and excretion. I solve the system using a daily Euler step that set the initial conditions $B(0) = S(0) = K(0) = 0$ and simulate lead levels from gestational day 0 to day 280. The resulting $B(t)$ gives the daily level of lead in maternal blood after accounting for the timing of race-day exposure, movement between compartments, and elimination. Because lead crosses the placenta readily and maternal and cord blood lead levels are highly correlated ([Zhu et al., 2010](#)), I use $B(t)$ as a proxy for fetal blood lead exposure throughout the analysis.

The bone-to-blood transfer rate, $k_{KB}(t)$, increases over pregnancy to reflect the increase in bone resorption documented in the toxicological literature:

$$k_{KB}(t) = \begin{cases} k_{KB}^{(1)} = 0.00045 & 0 \leq t < 90 \\ k_{KB}^{(2)} = 0.00080 & 90 \leq t < 180 \\ k_{KB}^{(3)} = 0.00150 & 180 \leq t \leq 280 \end{cases} \quad (10)$$

The above parameters are calibrated from the toxicological literature: $k_{B0} = 0.023$, $k_{BS} = 0.040$, $k_{SB} = 0.020$, $k_{BK} = 0.007$ (all per day) (O’Flaherty, 1991; U.S. Environmental Protection Agency, 2024). The bone to blood transfer rate increases from day 0 to day 280, with break points at time $t = 90$ and $t = 180$. Solving the PBPK system above produces a daily fetal blood lead for each mother i from conception to full term of pregnancy. I denote this trajectory $B_i(t)$ where t indexes the gestational day from $t = 1$ to $t = 280$ for each mother i . Since the 280 daily predicted fetal blood-lead concentrations for each mother i cannot directly enter the regression, I collapse each mother’s daily trajectory into three average concentrations, one for each trimester. Thus, each mother has an average predicted fetal blood-lead concentration for the first, second, and third trimesters. These three trimester-specific averages are then used as the exposure variables in the regression. Let T_k denote the gestational days in trimester $k \in \{1, 2, 3\}$, where $T_1 = \{0, \dots, 97\}$, $T_2 = \{98, \dots, 195\}$, and $T_3 = \{196, \dots, 280\}$. Let $|T_k|$ denote the number of gestational days in trimester k . The mean fetal blood lead for mother i in trimester k is then:

$$\bar{B}_{ik} = \frac{1}{|T_k|} \sum_{t \in T_k} B_i(t) \quad (11)$$

To measure cumulative fetal blood-lead exposure over pregnancy, I sum $B_i(t)$ across the 280 days of pregnancy to obtain the area under the trajectory (AUC):

$$\text{AUC}_i = \sum_{t=1}^{280} B_i(t). \quad (12)$$

Finally, the maximum daily fetal blood lead level during full pregnancy is given by,

$$\text{Peak}_i = \max_{t \in [1, 280]} B_i(t) \quad (13)$$

Hence, the trimester means $\bar{B}_{i1}, \bar{B}_{i2}, \bar{B}_{i3}$, the cumulative-dose measures AUC_i , and the

peak Peak_i describe the simulated fetal blood lead trajectory on when the exposure occurred, how much lead was accumulated overall, and how high it spiked at its worst is.

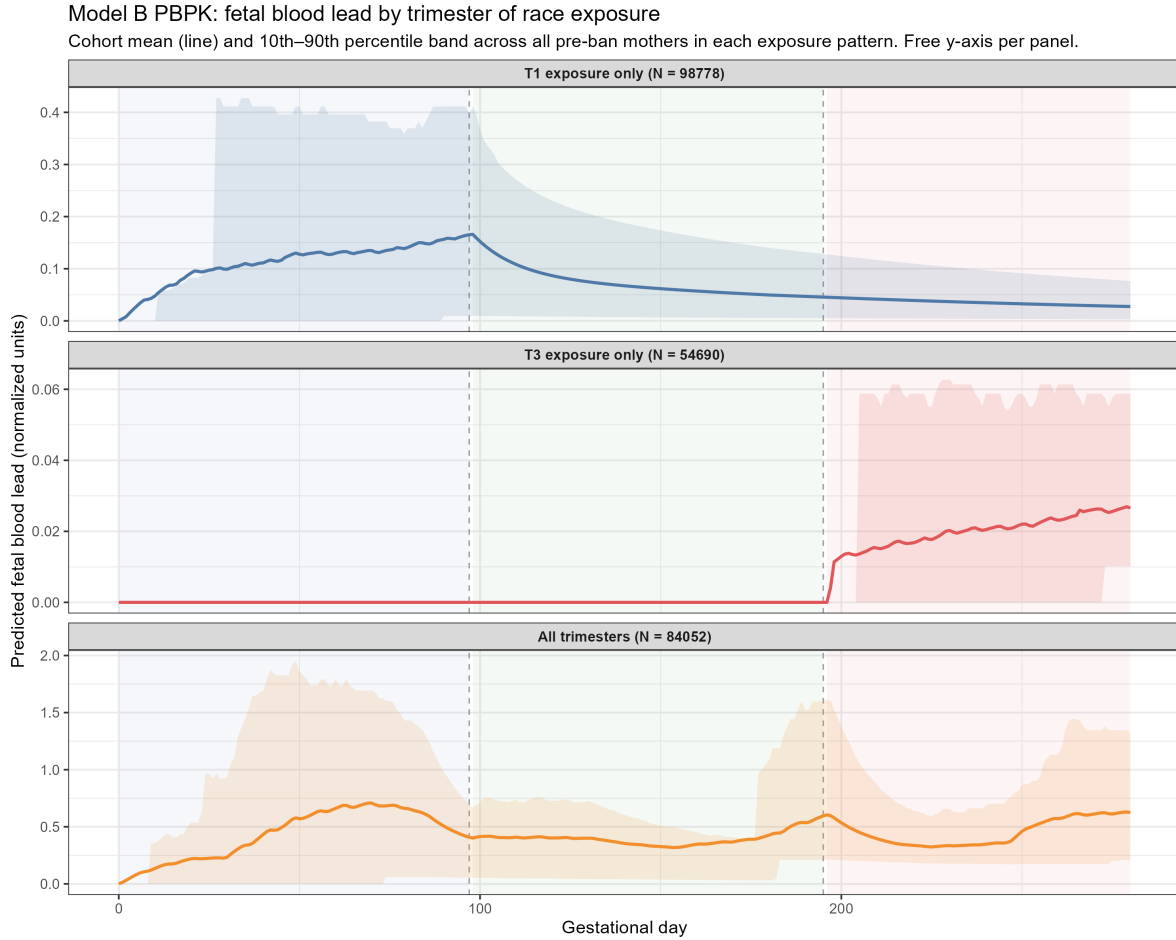


Figure 4. The top panel shows mothers whose only leaded race exposures occurred in the first trimester ($N = 98,778$); the middle panel shows mothers exposed only in the third trimester ($N = 54,690$); the bottom panel shows mothers exposed in all three trimesters ($N = 84,052$).

5 Empirical Strategy

5.1 Within-TMS Identification

The identification strategy relies on comparing births conceived before and after the 2007 switch to unleaded gasoline at TMS. The sample is restricted to all live births between 2003 to 2010 and who reside within 25 km of TMS. Conditional a mother's conception date and her residence, two mothers who conceive just a few weeks apart can be exposed to the same number of races but exposure can fall in different gestational windows. Additionally, because lead concentration decays with distance from the track, mothers who

live closer to the racetrack are more likely to be in higher lead plume than those living further away from the track. The spatial gradient test (Section 6.2) examines whether the effect on birthweight attenuates with distance from the track. Furthermore, two mothers can have very different exposures depending on whether their residence fall within the wind plume on race days even though they live the same equidistance to the racetrack. Finally, the level of emissions released by car engines varies depending on accidents and race length. These sources of variation are plausibly unrelated to maternal health. Appendix A tests whether log emissions are correlated with predetermined maternal characteristics such as maternal age, marital status, and race. The resulting coefficients are small and statistically indistinguishable from zero.

Since all mothers live within 25 km of TMS and experience at least one race during their pregnancy, exposure varies continuously across mothers depending on how many races fall in each trimester of their pregnancy. Hence, there is no control group in the traditional sense. Mothers with zero races in a given trimester provide the within-period baseline. Exploiting the timing of conception relative to the fixed calendar of TMS races, I estimate the effect of an additional race occurring in each trimester for pre and post ban periods. Since emissions measure is based on miles driven times grams per gallon (Equation 1), I can estimate the effect of lead on birth outcomes by comparing leaded race to “as if leaded” races. Given that race intensity continues to vary after 2007 even though the fuel no longer contains lead, we should expect the pre-ban coefficient to be negative and significant while post-ban coefficients to be close to zero. This would support the interpretation that the pre-ban effect is driven by lead rather than by other factors associated with racing.

5.2 Baseline Specification

The baseline reduced-form specification estimates the cross-sectional dose-response relationship separately in the pre-ban and post-ban samples:

$$BW_{ict} = \alpha + \sum_{k=1}^3 \beta_k Z_{ik} + X'_{ict} \gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \quad (14)$$

where BW_{ict} is birthweight in grams for birth i in census tract c at calendar time t ; Z_{ik} is the log trimester exposure regressors ($\tilde{E}_{ik}$ in Equation 3); X_{ict} is a vector of maternal demographic and socioeconomic controls (marital status, WIC enrollment, maternal risk indicator, education, race, parity, insurance status, and maternal age group); μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed effects.

Standard errors are clustered at the census tract level.³ I estimate Equation (14) separately for the pre-ban and post-ban samples using the same log emissions measure, $\tilde{E}_{ik}$. Because Equation (1) calculates emissions from miles driven and fuel economy without conditioning on fuel type, I can construct the measure in both periods. Before the ban, it measures the lead actually emitted during races. After the ban, it measures the amount of lead that would have been emitted if the race had used the same leaded fuel as before 2006. Thus, post-ban $\tilde{E}_{ik}$ is not a measure of actual lead exposure. Instead, it captures variation in race activity that would have generated different levels of lead emissions under the pre-ban fuel regime. The pre-ban β_k captures the relationship between lead exposure during trimester k and birthweight, while the post-ban β_k captures the relationship between this counterfactual measure and birthweight when races no longer emit lead. If the pre-ban relationship is driven by lead, the post-ban coefficient should be close to zero.

Similarly, the specification for wind direction is:

$$BW_{ict} = \alpha + \sum_{k=1}^3 \theta_k \text{ShareDownwind}_{ik} + \sum_{k=1}^3 \rho_k R_{ik} + X'_{ict} \gamma + \mu_c + \mu_y + \mu_m + \varepsilon_{ict} \quad (15)$$

where BW_{ict} is birthweight in grams for birth i in census tract c at calendar time t ; $\text{ShareDownwind}_{ik}$ is the fraction of trimester k race days on which mother i was downwind of TMS (Equation 4); R_{ik} is the number of races at TMS during trimester k of mother i 's pregnancy; θ_k is the effect of an additional unit of downwind share in trimester k ; ρ_k controls for the overall number of races in that trimester; X_{ict} is a vector of maternal demographic and socioeconomic controls (marital status, WIC enrollment, maternal risk indicator, education, race, parity, insurance status, and maternal age group); μ_c are census tract fixed effects; μ_y are birth-year fixed effects; and μ_m are birth-month fixed effects. Holding the number of races during the trimester fixed, if $\theta_k < 0$, mothers who were downwind on a larger share of those race days received greater plume exposure and should experience larger reductions in birthweight.

6 Results

This section presents the paper's central finding. If a mother is exposed to lead, the infant's birthweight is reduced by 0.047 gram for every 1% increase in emissions in her

³Birth-month fixed effects absorb seasonal patterns in birthweight (e.g., temperature during gestation, flu season) that could otherwise confound the seasonal NASCAR race schedule with seasonal determinants of birth outcomes. I verify that dropping birth-month fixed effects leaves the pre-ban coefficients virtually unchanged in magnitude, direction, and significance.

third trimester, holding the other trimesters constant. A doubling of emissions is associated with approximately a 3.23 ($-4.6564 \times \ln(2)$) gram decrease in birthweight. This effect is absent post-ban. A mother closer to the racetrack sees a larger birthweight reduction than a mother farther away during the pre-ban period, and this gap disappears post-ban.

6.1 Pre-Ban vs. Post-Ban Effect

Table 3 presents the main results for birthweight in grams, low birthweight (< 2500 grams), gestational age in weeks, and birthweight conditional on gestational age, with pre-ban and post-ban estimates reported side by side for each outcome. In Panel A, a one unit increase in third-trimester log emissions reduces birthweight by 4.65 grams pre-ban and is statistically insignificant post-ban. The first-trimester coefficient, by contrast, remains significant post-ban (-2.68). Third-trimester gestational age falls by .02 weeks pre-ban and rises by .01 weeks post-ban. In Panel B, each additional race lowers birthweight across all three trimesters pre-ban. In the post-ban period, trimesters 2 and 3 are null but trimester 1 remains marginally significant. The probability of low birthweight rises in trimesters 2 and 3 pre-ban while trimester 2 remains significant (.0075) post ban. Gestational age falls between .15 and .4 weeks across trimesters pre-ban, and trimester 3 again shows a marginal post-ban increase. The birthweight conditional on gestational age in both Panel A and Panel B show an inconsistent pattern such that we see a reduction in birthweight conditional on gestational age in trimester 1 (-3.17) in Panel A and (-16.78) in Panel B.

Table 3. Main Reduced-Form Results

Panel A: Log emissions (T1+T2+T3 joint)								
	Pre-ban (leaded)				Post-ban (as-if-leaded, placebo)			
	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA	(1) BW (g)	(2) LBW	(3) Gest. age (wks)	(4) BW GA
Log emissions, T1	2.1260 (1.4575)	-0.0015* (0.0008)	0.0297*** (0.0086)	-1.3368 (1.2860)	-2.6821** (1.2655)	0.0008 (0.0007)	0.0043 (0.0067)	-3.1702*** (1.0564)
Log emissions, T2	-3.1188 (2.1378)	0.0001 (0.0012)	-0.0255*** (0.0095)	-0.1425 (1.5365)	-0.6242 (1.4882)	0.0008 (0.0006)	-0.0056 (0.0064)	0.0015 (1.2600)
Log emissions, T3	-4.6564** (1.8932)	-0.0001 (0.0010)	-0.0175* (0.0088)	-2.6173 (1.6796)	0.3415 (1.3052)	-0.0001 (0.0006)	0.0122** (0.0058)	-1.0293 (1.2586)
Observations	15092	15092	15092	15092	28399	28399	28399	28399
R-squared	0.0340	0.0186	0.0415	0.2315	0.0328	0.0133	0.0259	0.2213
<i>Log emissions, full pregnancy</i>								
Log emissions, full pregnancy	-75.0806*** (11.7630)	0.0230*** (0.0060)	-0.5135*** (0.0660)	-15.3442 (9.9704)	-10.1609 (19.9455)	0.0030 (0.0099)	0.2424** (0.1071)	-37.4315** (16.8921)
Observations	15092	15092	15092	15092	28399	28399	28399	28399
R-squared	0.0357	0.0195	0.0450	0.2313	0.0326	0.0131	0.0258	0.2211
Panel B: Cumulative race count (T1+T2+T3 joint)								
	BW (g)	LBW	Gest. age (wks)	BW GA	BW (g)	LBW	Gest. age (wks)	BW GA
Races during T1	-25.3328*** (6.9562)	0.0056 (0.0035)	-0.1472*** (0.0384)	-8.1654 (4.9628)	-11.4191* (6.0972)	0.0043 (0.0031)	0.0476 (0.0296)	-16.7752*** (4.5863)
Races during T2	-48.6873*** (8.2176)	0.0162*** (0.0041)	-0.4048*** (0.0389)	-1.4754 (7.1483)	-7.8172 (7.1303)	0.0075** (0.0030)	-0.0021 (0.0324)	-7.5767 (6.0650)
Races during T3	-28.7844*** (7.3592)	0.0059* (0.0035)	-0.1971*** (0.0382)	-5.7968 (6.5800)	0.7409 (5.9614)	0.0021 (0.0029)	0.0503* (0.0281)	-4.9184 (5.5635)
Observations	15092	15092	15092	15092	28399	28399	28399	28399
<i>Cumulative race count, full pregnancy</i>								
Total Races, Full Pregnancy	-23.4470*** (5.8552)	0.0040 (0.0031)	-0.1334*** (0.0368)	-7.9113* (4.6024)	-4.3695 (4.7021)	0.0019 (0.0025)	0.0637** (0.0272)	-11.5378*** (4.0187)
Observations	15092	15092	15092	15092	28399	28399	28399	28399
R-squared	0.0344	0.0182	0.0397	0.2313	0.0327	0.0132	0.0258	0.2212

Standard errors in parentheses. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. Outcomes: birthweight in grams (BW), low birthweight <2500g (LBW), gestational age in weeks, and birthweight conditional on gestational age (BW | GA). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Spatial Gradient Test

If airborne lead is driving the relationship, birthweight should increase with distance from the track in the pre-ban period as lead concentrations decay with distance, while the birthweight gradient should be approximately flat in the post-ban period. Figure 5 shows that predicted birthweight rises with distance from the track during the pre-ban period. Intuitively, mothers who reside closer to TMS are exposed to higher concentrations of lead and deliver infants with lower birthweight. Once lead is removed from the fuel, we should expect to see the reverse such that the distance gradient flattens post-ban. This confirms that the pre-ban gradient was driven by lead rather than any other feature of living near a racetrack.

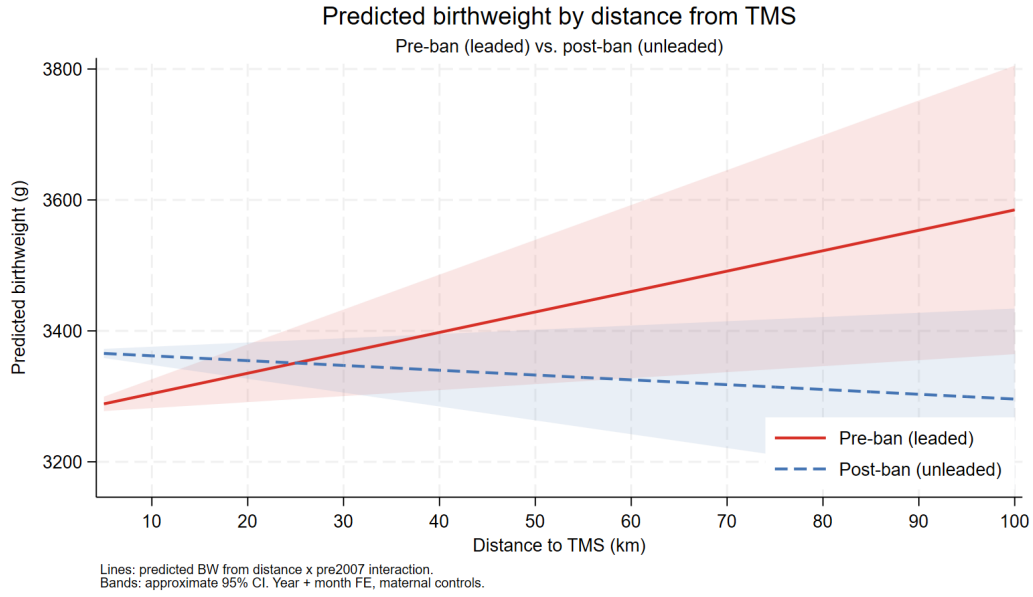


Figure 5. Predicted birthweight by distance from TMS. (Pre-ban = solid red; Post-ban = dashed blue). Birthweight is regressed on distance $\times$ pre-ban with year and month fixed effects and maternal controls. Shaded regions are approximate 95% confidence intervals.

The interaction coefficient (pre-ban $\times$ distance) in Figure 5 is 3.76 grams per kilometer and is statistically significant at the one percent level, while the post-ban distance slope is -0.78 grams per kilometer and is statistically insignificant. This implies that each additional kilometer from TMS is associated with an approximately 3-gram increase in birthweight during the leaded era, while the distance gradient is essentially flat in the post-ban period.

6.3 Wind Direction

Wind direction is plausibly exogenous to maternal sorting around the racetrack. Hence, mothers who reside downwind of the track on race days should experience larger birthweight reductions than those upwind, holding fixed the number of races during the trimester. Table 4 reports the effect of residing downwind on birthweight. Panel A uses the share of trimester k race days on which the mother was downwind, while Panel B uses a binary indicator for whether the mother was downwind on at least one race day during the trimester. In Panel A, the second and third trimester coefficients are nearly identical, -66.1^{**} and -65.9^{***} , respectively, both conditional on cumulative race count, and both are null post-ban. For two mothers exposed to the same number of third trimester races, a mother whose residence was downwind on all race days is estimated to deliver a baby approximately 66 grams lighter than a mother whose residence was never downwind on race days. In Panel B, a mother whose race days included at least one day with wind

blowing toward her residence sees birthweight fall by approximately 38.4 grams in the second trimester and 51.5 grams in the third trimester, relative to a mother whose race days never did, holding the number of races constant. In contrast, the result is null post-ban in both panels, except for the third trimester in Panel B, where the post-ban coefficient is significant and positive. This is an anomaly that does not fit the placebo pattern and is not explained by the current specification. Overall, if a mother resides downwind, she is expected to see a reduction in birthweight when lead is present.

Table 4. Wind Direction: Downwind Exposure and Birthweight

<i>Panel A: Share of race days downwind</i>		
	Pre-ban (1)	Post-ban (2)
Share downwind, T1	5.9628 (23.5948)	8.6250 (16.5578)
Share downwind, T2	-66.1419** (28.5564)	12.7569 (20.0366)
Share downwind, T3	-65.9353*** (21.1761)	19.9568 (14.4485)
Races during T1	-23.7100*** (7.1044)	-12.1012** (6.0173)
Races during T2	-47.1470*** (8.2486)	-8.7071 (7.6967)
Races during T3	-27.7798*** (7.9613)	-0.5022 (5.8915)
Observations	15092	28399
R-squared	0.0364	0.0329
<i>Panel B: Any downwind indicator</i>		
	Pre-ban	Post-ban
Any downwind, T1	-0.3420 (14.3705)	0.5701 (9.6261)
Any downwind, T2	-38.4413** (15.3719)	0.9807 (11.2574)
Any downwind, T3	-51.5110*** (14.4711)	21.1333*** (7.9341)
Races during T1	-24.0839*** (7.3524)	-11.4912* (6.0214)
Races during T2	-46.4629*** (7.9535)	-7.7781 (7.8623)
Races during T3	-24.9527*** (7.8124)	-2.1764 (6.0103)
Observations	15092	28399
R-squared	0.0364	0.0330

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects, standard maternal controls, and cumulative race counts (shown). Standard errors clustered by tract. The share downwind measure is the fraction of a trimester's race days on which the mother's residence was downwind ($\pm 45^\circ$). The any-downwind indicator equals one if the mother was downwind on at least one race day. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 PBPK Dose-Response

Sections 6.1 to 6.3 discuss the results from the reduced-form specification comparing the pre- and post-ban periods, and suggest that the reduction in birthweight operates through the lead channel, given that the results are statistically significant pre-ban and null post-ban. However, these reduced-form estimates do not capture the biological effects of lead in the maternal body. The PBPK specification translates these effects into biologically interpretable units by modeling how lead deposits and accumulates in blood, soft tissue, and bone, producing simulated fetal blood-lead trajectories. The PBPK regression is estimated on the pre-ban cross-section only, since blood lead is mechanically zero after the unleaded transition. Table 5 reports the results. In the joint specification (Column 4), a one-unit increase in mean fetal blood lead is associated with a birthweight decrease of 300 grams in the second trimester and 178 grams in the third trimester, both significant at the one percent level. Similarly, the cumulative impact of lead (AUC) over a mother's full pregnancy is -1.21 , and peak blood lead is associated with a birthweight decrease of 48.47 grams, significant at the one percent level. These coefficients represent the effect of a one-unit increase in PBPK-simulated blood lead. To put them in context, a one-standard-deviation increase in second-trimester blood lead implies a birthweight reduction of approximately $300 \times 0.21 \approx 63$ grams (mean = 0.317, interquartile range = 0.138 to 0.580). A mother at the 75th percentile of second-trimester blood lead delivers a baby roughly 133 grams lighter than a mother at the 25th percentile, holding all else fixed. Additionally, a one standard deviation increase in cumulative dose reduces birthweight by approximately 52 grams (1.23×42.33) while a one standard deviation increase in maximum blood lead reduces birthweight by approximately 45 grams (48.57×0.926). These confirm that mothers with higher cumulative or peak fetal blood lead deliver infants with lower birthweight.

Table 5. PBPK Dose-Response: Birthweight and Modeled Fetal Blood Lead (Pre-Ban)

	(1) T1 only	(2) T2 only	(3) T3 only	(4) Joint	(5) AUC	(6) Peak
Mean fetal blood lead, T1	-28.9799 (27.9391)			-25.2964 (26.8815)		
Mean fetal blood lead, T2		-240.1237*** (38.4502)		-299.6581*** (37.0532)		
Mean fetal blood lead, T3			-119.3553*** (36.0970)	-178.0472*** (35.4585)		
AUC, full pregnancy					-1.2095*** (0.2045)	
Peak blood lead						-48.5666*** (6.8134)
Observations	15092	15092	15092	15092	15092	15092
R-squared	0.0333	0.0363	0.0343	0.0388	0.0360	0.0358

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-ban only. All regressions include census tract and year and month fixed effects, plus standard maternal controls. Standard errors clustered by tract. (Table 3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 Mechanism: Maternal-Vascular Pathway

To investigate potential mechanisms, I re-estimate the joint exposure regression with alternative maternal and newborn outcomes on the left-hand side, reporting pre-ban and post-ban columns side by side (Table 6). The outcomes are gestational hypertension, eclampsia, five-minute Apgar score, and a low-Apgar indicator (<7). In Panel A, exposure to emission in the third trimester raises the probability of gestational hypertension (0.0017**) while the coefficient is small but significant and reverses sign post ban (-0.0007^*). In Panel B, each additional race in the third trimester increases the probability of gestational hypertension by about 1 percentage point pre-ban (0.0096) while the coefficient again reverses sign and remains significant, falling by about 0.3 percentage points in post ban period (-0.0032^{**}). Combined with the results in Table 3, a potential mechanism is that lead raises maternal blood pressure, which impairs placental circulation and, in turn, shortens gestation. Eclampsia, the five-minute Apgar score, and the low-Apgar indicator are all null post-ban in both panels, consistent with no detectable non-lead effect on these three outcomes. Hypertension is the exception as its post-ban coefficient remains significant and switches sign, which is the same pattern we see for gestational age and the wind-direction test.

Table 6. Mechanism: Maternal-Vascular and Newborn Outcomes

Panel A: Log emissions (T1+T2+T3 joint)

	Pre-ban (leaded)				Post-ban (as-if-leaded, placebo)			
	(1) Hyp.	(2) Ecl.	(3) Apgar5	(4) Low Apgar	(1) Hyp.	(2) Ecl.	(3) Apgar5	(4) Low Apgar
Log emissions, T1	-0.0013** (0.0006)	0.0001 (0.0001)	-0.0029* (0.0016)	0.0005* (0.0003)	0.0003 (0.0003)	-0.0000 (0.0001)	-0.0017 (0.0012)	0.0002 (0.0002)
Log emissions, T2	-0.0001 (0.0008)	-0.0000 (0.0001)	-0.0030* (0.0018)	0.0003 (0.0003)	-0.0007 (0.0004)	-0.0000 (0.0001)	-0.0008 (0.0015)	-0.0001 (0.0002)
Log emissions, T3	0.0017** (0.0007)	0.0001 (0.0002)	0.0020 (0.0021)	-0.0002 (0.0003)	-0.0007* (0.0004)	-0.0000 (0.0001)	-0.0002 (0.0015)	-0.0001 (0.0003)
Observations	15092	15092	14565	14565	28399	28399	27688	27688
Log emissions, full pregnancy	0.0115*** (0.0043)	0.0017 (0.0012)	0.0041 (0.0136)	-0.0004 (0.0026)	-0.0067 (0.0056)	0.0002 (0.0009)	-0.0086 (0.0272)	-0.0001 (0.0055)
Observations	15092	15092	14565	14565	28399	28399	27688	27688
R-squared	0.2162	0.0125	0.0120	0.0064	0.1880	0.0069	0.0130	0.0059

Panel B: Cumulative race count (T1+T2+T3 joint)

	Hyp.	Ecl.	Apgar5	Low Apgar	Hyp.	Ecl.	Apgar5	Low Apgar
Races during T1	-0.0015 (0.0028)	0.0013** (0.0005)	-0.0172* (0.0095)	0.0031* (0.0016)	0.0002 (0.0020)	-0.0001 (0.0003)	-0.0077 (0.0075)	0.0009 (0.0013)
Races during T2	0.0056 (0.0036)	0.0010 (0.0006)	-0.0120 (0.0107)	0.0014 (0.0016)	-0.0037 (0.0022)	-0.0002 (0.0003)	-0.0033 (0.0075)	-0.0001 (0.0012)
Races during T3	0.0096*** (0.0028)	0.0009 (0.0008)	0.0122 (0.0079)	-0.0014 (0.0015)	-0.0032** (0.0016)	0.0000 (0.0003)	0.0013 (0.0065)	-0.0005 (0.0014)
Observations	15092	15092	14565	14565	28399	28399	27688	27688
Cumulative race count, full pregnancy	0.0039* (0.0023)	0.0011* (0.0006)	-0.0005 (0.0070)	0.0007 (0.0013)	-0.0010 (0.0014)	0.0000 (0.0002)	-0.0030 (0.0065)	0.0003 (0.0014)
Observations	15092	15092	14565	14565	28399	28399	27688	27688
R-squared	0.2161	0.0129	0.0120	0.0064	0.1879	0.0069	0.0130	0.0059

Standard errors in parentheses. Outcomes (LPM for binary): gestational hypertension (Hyp.), eclampsia (Ecl.), Apgar 5-min score, low Apgar indicator (<7). Sample: TMS live births within 25 km. All regressions include census tract and year and month fixed effects plus standard maternal controls. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

This study provides causal evidence that even low levels of background lead exposure can adversely affect maternal and fetal health, including in environments with relatively low background pollution. Using NASCAR's and ARCA's transition from leaded to unleaded gasoline at TMS in 2007, I find that prenatal exposure to leaded race days reduces birthweight, with a potential mechanism pointing to a maternal vascular hypertension pathway. The effects are mostly concentrated in the second and third trimesters, attenuate with distance from the track, and are amplified for mothers residing downwind on race days. Once lead is removed from the fuel, the effect disappears. Identifying the effects entirely from within-track variation at TMS, I show that lead reduces birthweight

through variation in pregnancy timing, distance, and wind direction, making the results difficult to reconcile with alternative explanations such as traffic congestion or maternal sorting. The reduced-form estimates, along with the PBPK-simulated trajectories, provide reassurance that the results are not artifacts of modeling assumptions or driven by non-lead channels. Furthermore, the Appendix examines EPA monitors that detect lead readings, and the results are consistent with the PBPK results. More broadly, the results suggest that even niche pollution sources can generate measurable public health costs and that targeted environmental regulations yield substantial benefits when they eliminate highly concentrated sources of toxic exposure. The 2007 NASCAR fuel transition was a small policy at the national scale, and this research shows that it had measurable effects on the fetal health of mothers living near a single venue.

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A Balance Test

Table 7 reports the balance test referenced in Section 5.1. Each column is a separate regression of pre-determined maternal characteristics on joint log emissions with census tract and year and month fixed effects on the pre-ban sample.

Table 7. Log Emissions vs. Maternal Characteristics (Pre-Ban)

	(1) Mat. age	(2) Married	(3) WIC	(4) Black	(5) White	(6) First birth	(7) Insured
Log emissions, T1	0.0007 (0.0118)	-0.0002 (0.0007)	0.0013 (0.0013)	-0.0004 (0.0004)	0.0008 (0.0007)	0.0010 (0.0009)	-0.0003 (0.0008)
Log emissions, T2	0.0075 (0.0122)	0.0003 (0.0009)	0.0017 (0.0015)	0.0002 (0.0005)	0.0008 (0.0007)	0.0008 (0.0008)	0.0007 (0.0010)
Log emissions, T3	0.0002 (0.0115)	0.0007 (0.0006)	0.0002 (0.0009)	0.0001 (0.0005)	0.0005 (0.0008)	-0.0008 (0.0009)	0.0002 (0.0009)
Observations	31141	31141	15092	31141	31141	31141	15092
R-squared	0.1345	0.0899	0.1653	0.0248	0.0332	0.0105	0.1060

Standard errors in parentheses. Each column is a separate regression of the indicated pre-determined maternal characteristic on joint log emissions (T1+T2+T3) during pregnancy. Sample: TMS live births within 25 km, pre-ban (conception before Dec 17, 2005). Fixed effects: census tract and year-by-month of conception. Standard errors clustered by tract. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Monitor-Based Exposure

For each race day, I compute an inverse distance weighted (IDW) ambient lead concentration at each mother's residence using all EPA monitors within 150 km:

$$w_{im} = \frac{1}{\max(d_{im}, 1)}, \quad \text{Lead}_{ij} = \frac{\sum_m w_{im} \cdot \text{ExcessLead}_{mj}}{\sum_m w_{im}} \quad (16)$$

where w_{im} is the inverse-distance weight assigned to monitor m for mother i ; d_{im} is the distance in kilometers between mother i 's census tract centroid and monitor m (monitors beyond 150 km receive zero weight); ExcessLead_{mj} is the race-day excess lead concentration at monitor m on race day j (the monitor's reading minus its annual non-race-day baseline); and Lead_{ij} is the IDW ambient lead concentration assigned to mother i on race day j measured in $\mu\text{g}/\text{m}^3$.

The race-day absorbed dose for mother i on race day j is then:

$$q_{ij}^A = a \cdot \text{Lead}_{ij} \cdot IR_i \cdot \Delta t_{ij} \quad (17)$$

where q_{ij} is the absorbed lead dose (measured in micrograms) for mother i on race day j ; Lead_{ij} is the IDW ambient lead concentration from Equation 16; $a = 0.3$ is the standard pulmonary absorption fraction (Oflaherty, 1993); $IR_i = 16 \text{ m}^3/\text{day}$ is the maternal inhalation rate (U.S. Environmental Protection Agency, 2011); and $\Delta t = 1 \text{ day}$ reflects the 24-hour averaging window of the monitor reading. Mothers with no monitor coverage during the trimester receive a dose of zero.

Table 8. EPA AQS lead monitors at TMS: race-day coverage 2003–2010

Site ID	County	Distance	Years active	Total readings	Race days (pre)	Race days (post)	Mean excess Pb	Consistent
480850009	Collin	43.7	2003–2010	469	3	3	0.000	✓
480850003	Collin	44.0	2003–2010	463	3	3	0.000	✓
480850007	Collin	44.0	2003–2010	463	3	3	0.000	✓
481130057	Dallas	47.8	2003–2007	1393	14	0	0.023	✓
481130018	Dallas	56.9	2003–2007	239	3	0	0.024	✓
481130061	Dallas	47.1	2003–2003	8	0	0	–	
481130066	Dallas	57.2	2003–2004	101	1	0	0.046	

Notes: These EPA Air Quality System (AQS) ambient lead monitors are within $\approx 161 \text{ km}$ of Texas Motor Speedway with at least one valid reading during 2003–2010. Distance is the distance from the monitor’s reported coordinates to TMS. “Years active” is the year in which the monitor reported any reading, regardless of whether the reading fell on a race day. “Total readings” is the count of all valid daily readings the monitor reported over 2003–2010.

“Race days (pre)” counts the number of TMS race days on which this monitor reported a reading restricted to race days through Dec. 17, 2005; “Race days (post)” covers race days from Mar. 1, 2007 onward. “Mean excess Pb” is the monitor’s mean race-day reading minus its annual non-race-day baseline, averaged over its pre-ban race day (in $\mu\text{g}/\text{m}^3$); “–” indicates the monitor recorded no pre-ban race-day readings.

“Consistent” comprises monitors that operated continuously throughout the pre-ban window (2003–2006) and is used in the robustness check reported in Section ?? . Site 481130066 is excluded because it dropped out after 2004. Site 481130061 is excluded because it operated only briefly in 2003 and caught no race days. Of the 17 pre-ban race-day readings in the consistent set, 14 come from site 481130057 in Dallas County.

B.1 Monitor Calibration

Table 9 reports the results in three panels. Panel A reports the absorbed dose by trimester. In the joint specification, the first and second trimesters have the largest effects. Similarly, Panel B reports the PBPK-simulated trimester-mean blood lead alongside the full-pregnancy cumulative dose (AUC), and the first and second trimesters dominate which is different from Table 5, where the second and third trimesters dominate. Panel C reports the cumulative race count by trimester and the full-pregnancy total. While the results differ by trimester, the monitor-based exposure and emission-based exposure suggest that higher prenatal lead exposure reduces birthweight.

Table 9. Model A Calibration: Monitor-Based Exposure (Pre-Ban)

<i>Panel A: Absorbed dose (monitor-based)</i>				
	(1) T1 only	(2) T2 only	(3) T3 only	(4) Joint
Absorbed dose, T1	-55.9466*** (18.7717)			-92.9875*** (21.3191)
Absorbed dose, T2		-34.2125** (16.7463)		-71.7626*** (19.1848)
Absorbed dose, T3			-21.9377 (25.6915)	-41.3616 (26.6056)
Observations	15092	15092	15092	15092
R-squared	0.0337	0.0335	0.0333	0.0348
<i>Panel B: PBPK trimester means and cumulative dose</i>				
	T1 only	T2 only	T3 only	Joint
Mean fetal blood lead, T1	-349.1209*** (115.9916)			-364.1708*** (114.6612)
Mean fetal blood lead, T2		-486.7370*** (131.9287)		-439.1981*** (140.1733)
Mean fetal blood lead, T3			-285.9910** (120.7131)	-167.1096 (126.9874)
Observations	15092	15092	15092	15092
Cumulative dose (AUC)	-3.2723*** (0.6851)			
Observations	15092			
R-squared	0.0351			
<i>Panel C: Cumulative race count</i>				
	T1 only	T2 only	T3 only	Joint
Races during T1	0.0978 (5.6670)			-25.3328*** (6.9562)
Races during T2		-17.3989*** (5.8493)		-48.6873*** (8.2176)
Races during T3			-5.8222 (6.3340)	-28.7844*** (7.3592)
Observations	15092	15092	15092	15092
Races, full pregnancy	-23.4470*** (5.8552)			
Observations	15092			
R-squared	0.0344			

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-ban only. Fixed effects: census tract, year, month. Standard errors clustered by tract.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Consistent-Monitor Robustness

A potential concern with using EPA monitors is that EPA monitor coverage near TMS changed over the sample period. Of the seven monitors used in the monitor-based specification, two dropped out during the pre-ban window (site 481130066 after 2004 and site 481130061 after 2003). This could make the inference is fragile. To address monitor attrition, I re-estimate the model's regressions using only the five monitors active throughout the entire pre-ban window (2003–2006) and compare the results side by side. (Table 10) reports the birthweight in grams.

In Panel A, the full-monitor coverage produces coefficient of -92.9 while the consistent-monitor coverage produces coefficient is -93.7 in the first trimester, all significant at the one percent level. This is the same for the rest of the trimesters. In Panel B, the full-monitor coverage produces a coefficient of -439.20 while the consistent-monitor coefficient is -441.55 in the second trimester, again both significant at the one percent level. In Panel C, the full-monitor coefficient is -3.27 and the consistent-monitor coefficient is -3.29 , both significant at the one percent level. The standard errors are nearly identical across specifications. This stability rules out monitor attrition as a driver of the model's result.

Table 10. Consistent-Monitor Robustness

<i>Panel A: Absorbed dose (joint, pre-ban)</i>		
	(1) Full (7 monitors)	(2) Consistent (5 monitors)
Absorbed dose, T1	-92.9875*** (21.3191)	-93.7305*** (21.3241)
Absorbed dose, T2	-71.7626*** (19.1848)	-72.0242*** (19.1859)
Absorbed dose, T3	-41.3616 (26.6056)	-41.3776 (26.6009)
Observations	15092	15092
R-squared	0.0348	0.0348
<i>Panel B: PBPK trimester means (joint, pre-ban)</i>		
	Full (7 monitors)	Consistent (5 monitors)
Mean fetal blood lead, T1	-364.1708*** (114.6612)	-368.8793*** (114.6463)
Mean fetal blood lead, T2	-439.1981*** (140.1733)	-441.5516*** (140.2599)
Mean fetal blood lead, T3	-167.1096 (126.9874)	-166.3257 (126.9548)
Observations	15092	15092
R-squared	0.0352	0.0352
<i>Panel C: Cumulative dose (pre-ban)</i>		
	Full (7 monitors)	Consistent (5 monitors)
Cumulative dose (AUC)	-3.2723*** (0.6851)	-3.2893*** (0.6855)
Observations	15092	15092
R-squared	0.0351	0.0351

Standard errors in parentheses. Outcome: birthweight in grams. Sample: TMS live births within 25 km, pre-ban only. Fixed effects: census tract, year, month. Standard errors clustered by tract. Full (7 monitors) uses 7 EPA monitors within 100 km of TMS with race-day coverage during 2003–2010. Consistent (5 monitors) restricts to the 5 monitors active throughout 2003–2006. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Threats to Cross-Track Comparisons

NASCAR and ARCA also held races at venues other than TMS. Thunder Hill Raceway hosted five regional races between 2005 and 2009, while Meyer Speedway and Texas World Speedway are legacy racetracks with no NASCAR activity during the sample period. Table 11 examines whether using these venues in a cross-track comparison would af-

fect identification relative to the within-TMS design. Panel A reports the results for Thunder Hill Raceway. Exposure to a leaded race is associated with a birthweight reduction of about 17 grams in the third trimester, while the first-trimester estimates are positive in both the pre- and post-ban periods. However, Thunder Hill hosted only five regional races, two of which were leaded, making it an unreliable comparison site for identifying the effects of lead exposure. In Panel B, Meyer Speedway exhibits a negative and statistically significant post-2007 coefficient of approximately 25 grams, whereas Texas World Speedway shows no significant change. The decline in birthweight at Meyer Speedway may reflect a shock in the Houston area after 2007 rather than NASCAR-related factors. It could potentially be driven by local demographic and economic changes around the 2008 financial crisis. Given that there is no similar decline at Texas World Speedway, the change may be specific to the Houston area rather than reflecting a general effect of lead exposure. These results highlight a potential problem with using births near Meyer as a control group for those near TMS. The within-TMS design avoids this issue by restricting the sample to mothers living near TMS and exploiting variation in race exposure within census tracts and across birth years and months. This approach addresses threats to cross-track comparisons that may be affected by different regional trends.

Table 11. Replication and Falsification at Other Texas Tracks

<i>Panel A: Thunder Hill — log emissions (pre — post)</i>		
	(1) Pre-ban	(2) Post-ban (as-if-leaded, placebo)
Log emissions, T1	9.8134* (4.9643)	9.2335*** (1.7574)
Log emissions, T2	4.7616 (3.3354)	6.9178* (3.6090)
Log emissions, T3	-17.1740*** (5.6452)	5.8880 (3.7305)
Observations	4241	8986
R-squared	0.0542	0.0371
Log emissions, full pregnancy	1.0641 (2.7738)	7.6203*** (2.0941)
Observations	4241	8986
R-squared	0.0515	0.0370
<i>Panel A (cont.): Thunder Hill — cumulative race count (pre + post)</i>		
	Pre-ban	Post-ban (placebo)
Races during T1	69.3993* (35.1073)	64.7881*** (12.4589)
Races during T2	33.6736 (23.5875)	48.2251* (25.7843)
Races during T3	-121.4530*** (39.9225)	41.5483 (26.1554)
Observations	4241	8986
R-squared	0.0542	0.0371
Total Races, Full Pregnancy	7.5249 (19.6164)	53.3642*** (15.1547)
Observations	4241	8986
R-squared	0.0515	0.0370
<i>Panel B: Post-2007 birthweight break at legacy tracks (falsification)</i>		
	Meyer (Houston)	TX World (College Stn.)
Post-ban	-25.2847*** (4.6394)	2.5829 (17.5699)
Year trend	4.5016*** (1.3251)	-3.1908 (5.2259)
Observations	215615	12332
R-squared	0.0418	0.0416

Standard errors in parentheses. Outcome: birthweight in grams. City fixed effects due to the small geographic area. Sample: births within 25 km. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Race Schedule

Table 12 lists every NASCAR-sanctioned race at the four Texas tracks during the sample period (2003–2010). TMS hosted multiple national-series races each year, typically clustering around two weekends in April and November. Thunder Hill hosted a small number of KNPSW regional races between 2005 and 2009. Texas World Speedway and Meyer Speedway had no races during the sample period and are omitted.

Table 12. NASCAR and ARCA Race Schedule at Texas Tracks, 2003–2010

Track	Date	Series	Fuel	Miles	Drivers
Thunder Hill	10/01/2005	KNPSW	Leaded	1132	20
Thunder Hill	09/30/2006	KNPSW	Leaded	1156	22
Thunder Hill	03/31/2007	KNPSW	Unleaded	1472	28
Thunder Hill	04/19/2008	KNPSW	Unleaded	1188	22
Thunder Hill	03/14/2009	KNPSW	Unleaded	974	19
TMS	03/29/2003	Xfinity	Leaded	10236	43
TMS	03/30/2003	Cup	Leaded	19646	43
TMS	06/06/2003	CWTS	Leaded	6538	36
TMS	10/11/2003	CWTS	Leaded	6537	36
TMS	04/03/2004	Xfinity	Leaded	10576	43
TMS	04/04/2004	Cup	Leaded	19016	43
TMS	06/11/2004	CWTS	Leaded	6651	34
TMS	10/16/2004	CWTS	Leaded	6568	36
TMS	04/16/2005	Xfinity	Leaded	11724	43
TMS	04/17/2005	Cup	Leaded	19006	43
TMS	06/10/2005	CWTS	Leaded	8067	36
TMS	11/04/2005	CWTS	Leaded	7262	36
TMS	11/05/2005	Xfinity	Leaded	11922	43
TMS	11/06/2005	Cup	Leaded	20862	43
TMS	04/08/2006	Xfinity	Leaded	12855	43
TMS	04/09/2006	Cup	Leaded	19456	43
TMS	06/09/2006	CWTS	Leaded	8409	36
TMS	11/03/2006	CWTS	Unleaded	7240	36
TMS	11/04/2006	Xfinity	Unleaded	11392	43
TMS	11/05/2006	Cup	Leaded	20432	43
TMS	04/14/2007	Xfinity	Unleaded	10738	43
TMS	04/15/2007	Cup	Unleaded	19338	43
TMS	06/08/2007	CWTS	Unleaded	6768	35
TMS	11/02/2007	CWTS	Unleaded	6540	36
TMS	11/03/2007	Xfinity	Unleaded	11133	43
TMS	11/04/2007	Cup	Unleaded	19131	43
TMS	04/05/2008	Xfinity	Unleaded	10794	43
TMS	04/06/2008	Cup	Unleaded	20853	43
TMS	06/06/2008	CWTS	Unleaded	7508	35
TMS	10/31/2008	CWTS	Unleaded	5216	34
TMS	11/01/2008	Xfinity	Unleaded	10635	43
TMS	11/02/2008	Cup	Unleaded	21098	43
TMS	04/04/2009	Xfinity	Unleaded	11266	43
TMS	04/05/2009	Cup	Unleaded	19972	43
TMS	06/05/2009	CWTS	Unleaded	5848	33
TMS	11/06/2009	CWTS	Unleaded	5968	36
TMS	11/07/2009	Xfinity	Unleaded	8836	43
TMS	11/08/2009	Cup	Unleaded	18784	43
TMS	04/16/2010	Arca	Unleaded	4350	38
TMS	04/19/2010	Cup	Unleaded	19040	43
TMS	04/19/2010	Xfinity	Unleaded	10138	43
TMS	06/04/2010	CWTS	Unleaded	5962	36
TMS	11/05/2010	CWTS	Unleaded	6591	36
TMS	11/06/2010	Xfinity	Unleaded	10760	43
TMS	11/07/2010	Cup	Unleaded	18435	43

All NASCAR- and ARCA-sanctioned races at the four Texas tracks during the sample period (2003–2010). TMS hosted Cup Series, Xfinity Series, and Camping World Truck Series races in most years, typically clustering around two weekends per year (April and November). Thunder Hill hosted five KNPSW regional races between 2005 and 2009. Texas World Speedway and Meyer Speedway hosted no races during the sample period (last active in 1993 and 1971, respectively) and are omitted from this table.